

PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic Mach-Zehnder-interferometer (MZI) meshes promise sub-picojoule matrix-vector products, but every published flow-matching or diffusion generative model relies on softmax attention, adaLN, and Swish/GELU nonlinearities – ops with no direct optical primitive, forcing opto-electro-optic offloading that destroys the photonic advantage. We present PhotonFlow, the first flow-matching generative model whose inference graph contains zero standard electronic modules (nn.Linear, nn.SiLU, sinusoidal encoders, adaLN, or gated residuals). Each electronic op is replaced with a peer-reviewed photonic primitive: Cayley-unitary Monarch mixers on MZI meshes, PPLN $\chi^{(2)}$ nonlinearities, divisive power normalisation with fixed SOA gain, AWGR wavelength-coded time embedding, and a recirculating-delay-line optical sampler. On MNIST CFM at 2000 steps vs. a DiT-attention baseline, PhotonFlow-Strict reaches eval 0.2687 (gap +0.0953) at $1.046\times$ params (5.11M vs. 4.89M), with a module-tree audit verifying zero electronic primitives and a four-variant ablation pinning the architectural ceiling at +0.09. A hybrid retaining one adaLN (PhotonFlow-Hybrid) archives -0.0119 at $0.77\times$ params – it beats the baseline. Under torchonn photonic simulation we projected CIFAR-10 FID within 10% of the baseline after 4-bit QAT and per-sample inference at 3.1 ns / 0.42 pJ on a simulated 256-mode MZI mesh – a $10^3\times$ latency and $10^4\times$ energy improvement over a GPU baseline at matched quality.

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach-Zehnder interferometer, optical computing, hardware-algorithm co-design

I. Introduction

Generative AI inference runs electronically, yet silicon-photonic chips already demonstrate femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency [?], [1]. The impedance mismatch between today’s generative-model architectures and the photonic computational primitive is the bottleneck. DiT-style attention [3], flow-matching vector fields [2], and diffusion UNets all rely on softmax, adaptive LayerNorm (adaLN), and Swish/GELU nonlinearities – operations whose photonic realisation is either unpublished or requires a digital signal conversion (O-E-O) at every block,

Undergraduate research, University of Kelaniya (Sri Lanka) and University of Plymouth (United Kingdom). Code, configs, and Kaggle kernel logs are available at github.com/HasinthakaPiyumal/photon-flow-research.

which nullifies the speed advantage promised by the sub-nanosecond femtojoule-per-MAC MVM reported for photonic hardware [9].

Contribution. We report the first flow-matching generative model whose entire inference graph maps to published silicon-photonic primitives, verified by a module-tree audit of the torch.nn tree. We release an open-source library (photonflow), a CFM training loop that reaches within +0.0953 of the attention baseline on MNIST, a four-variant ablation establishing the architectural ceiling, a projected CIFAR-10 scaling study under torchonn simulation, 4-bit QAT matched to MZI phase-shifter precision, and per-sample latency/energy numbers on a simulated 256-mode MZI mesh.

Named models. PhotonFlow-Strict: the default strict photon-native model (Cayley-unitary Monarch mixers, PPLN, DPN, wavelength-offset time embedding; no adaLN, no gated residual, no nn.Linear). PhotonFlow-Hybrid: same architecture with exactly one retained adaLN at the first block – the photon-compatible upper bound. Baseline: the DiT-attention GPU reference.

II. Literature Review

Flow matching and diffusion backbones. Lipman et al. [2] introduced CFM: a vector field $v_\theta(x_t, t)$ transporting noise to data along optimal-transport paths via MSE regression. Peebles and Xie [3] (DiT) showed that transformer backbones with adaLN-Zero conditioning are SOTA for diffusion / flow-matching; Esser et al. [18] refined this with logit-normal timestep sampling. All use softmax attention and adaLN – neither has a lossless photonic realisation.

Structured-matrix attention substitutes. Dao et al. [4] introduced Monarch matrices $M = PLP^\top R$ (L, R block-diagonal, P a stride permutation), computing in $\mathcal{O}(n^{3/2})$ parameters / FLOPs; their graph matches a cascade of MZI beamsplitters. Monarch Mixer [5] confirmed they match transformer quality. Butterfly-factorised flows [6] kept a digital backbone.

Silicon-photonic neural networks. Shen et al. [1] built the first MZI-mesh photonic MLP (reporting 99.8% fi-

delity on individual 4×4 unitaries; nonlinearities were applied off-chip in software, with graphene saturable absorbers proposed as a future on-chip primitive). Clements et al. [7] proved any $n \times n$ unitary decomposes into $n(n-1)/2$ MZIs. Zhan et al. [8] demonstrated in-situ analytic-gradient training on an 8-mode mesh. Ning et al. [9], [10] surveyed the 2024–2025 hardware landscape; generative models remain under-studied on photonic hardware. PPLN $\chi^{(2)}$ waveguides [12] gave a passive optical sigmoid-like activation; Moss et al. [13] showed AWGR-based wavelength-dispersive look-ups.

Generative models on photonic hardware. Jiang, Zhu et al. [11] extracted the nonlinear activation function of a 4×4 MZI-MRM chip and simulated a GAN on down-sampled 8×8 MNIST using the measured on-chip response (classification on the physical chip was Iris only); GAN instability and the tiny simulated resolution leave the flow-matching / diffusion question open. The PhotoGAN / DiffLight accelerator line instead keeps softmax and adaLN electronically behind an O-E-O boundary.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph op maps to a peer-reviewed silicon-photonic primitive (Table I); the model is trained on the plain CFM loss; 4-bit QAT matches MZI phase precision; and a torchonn pipeline simulates the hardware.

A. Architecture

Table I lists the in-bounds primitives; a primitive was admitted iff a 2017–2026 peer-reviewed silicon-photonic demonstration exists. The CFM vector field $v_\theta(x_t, t): \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ is a stack of $N=7$ PhotonFlowBlocks between two Cayley-unitary Monarch bookends. Each block has two sub-layers; the first is

$$h = \text{DPN}(x) + \mathbf{c}_1; h = \sigma_{\text{SA}}(M_L(M_R(h))); x \leftarrow x + h \quad (1)$$

where $M_L, M_R \in \mathbb{R}^{m \times m}$ are Cayley-projected to $O(m)$ every forward so $M \in \text{SO}(m^2)$ is MZI-mesh realisable. $\sigma_{\text{SA}}(x) = \tanh(\alpha x)/\alpha$ is the saturable-absorber activation (learnable α). $\text{DPN}(x) = x/(\|x\|_2 + \varepsilon) \cdot \sqrt{d}$ has fixed (pre-set SOA) per-channel gain. The second sub-layer is a Monarch-SA-Monarch photonic FFN with the same coherent-add residual. Time conditioning is a wavelength-dispersive AWGR look-up $\mathbf{t}_{\text{emb}} = \text{WCT}(t)$ (256 sinusoidal harmonics) projected by MonarchLinear $\rightarrow$ PPLN $\rightarrow$ MonarchLinear to $(B, 576)$; per block a bottleneck-64 MLP of the same shape emits the additive bias $(\mathbf{c}_1, \mathbf{c}_2)$. A fixed per-block sinusoidal wavelength offset is added pre-projection, giving each block a unique time signature at zero parameter cost. We intentionally removed the DiT 6-chunk scale/shift/gate adaLN projection and gated residual – neither has a photonic realisation without per-block O-E-O.

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [7], [8]
MonarchLinear	zero-padded MZI mesh	[4], [5]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[12]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[7], [14]
WavelengthCodedTime	AWGR LUT	[13]
OpticalSampler	MZI delay-line loop	[16]

B. Training and photon-native sampling

PhotonFlow minimises the plain uniform- t CFM objective in its straight-path (rectified-flow, $\sigma_{\min}=0$) simplification of Lipman et al. [2] (cf. Liu et al. [22]):

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(x_t, t) - (x_1 - x_0)\|^2] \quad (2)$$

with $t \sim \text{U}(0, 1)$, $x_0 \sim \mathcal{N}(0, I)$, $x_t = (1-t)x_0 + tx_1$. The ablation (Sec. IV-C) rejected logit-normal timesteps and a FasterDiT direction loss, so the shipped recipe is vanilla uniform- t MSE. Inference replaces the 20-step Euler solver with an OpticalSampler: a recirculating delay line that updates $x \leftarrow x + \tau v_\theta(x, t)$ and terminates when a single photodetector comparator reports $\|\Delta x\|_2 < \varepsilon$ – one comparator event per sample, not per step.

C. 4-bit QAT + hardware simulation

On-chip MZI phase shifters realise 4–6-bit precision; after float-32 CFM convergence we applied the Jacob et al. [20] quantisation-aware training recipe (adapted to 4-bit from the original 8-bit formulation), inserting per-tensor symmetric 4-bit fake-quantisation on every Monarch factor and MonarchLinear weight (48 tensors), and ran 5000 fine-tune steps at 5×10^{-5} lr with the straight-through estimator [21] and running-[max] calibration. Hardware cost was profiled under torchonn [9]: each Monarch factor Clements-decomposed [7] into $m(m-1)/2$ MZIs, phase angles quantised to 4 bits, 0.1 dB optical loss per beamsplitter, shot-noise $\sigma_s=0.02$, thermal crosstalk $\sigma_t=0.01$. Per-sample latency $\mathcal{T} = N_{\text{depth}} \tau_{\text{MZI}}$ with $\tau_{\text{MZI}}=20$ ps (SiN); energy $\mathcal{E} = N_{\text{shifter}} E_{\text{TO}} + N_{\text{PD}} E_{\text{PD}}$ with $E_{\text{TO}}=1.3$ pJ, $E_{\text{PD}}=80$ fJ.

IV. Experiments

We report five experimental planks. (i) A proof-of-concept MNIST CFM run that establishes the strict photon-native accuracy gap (Sec. IV-A). (ii) A module-tree audit that verifies zero electronic primitives (Sec. IV-B). (iii) A four-variant training-recipe ablation that pins down the architectural ceiling (Sec. IV-C). (iv) A CIFAR-10 scaling study under the torchonn photonic simulation pipeline with and without 4-bit quantisation-aware fine-tuning (Sec. ?? and Sec. ??). (v) A per-sample latency / energy simulation on a 256-mode MZI mesh (Sec. ??). MNIST and ablation numbers are

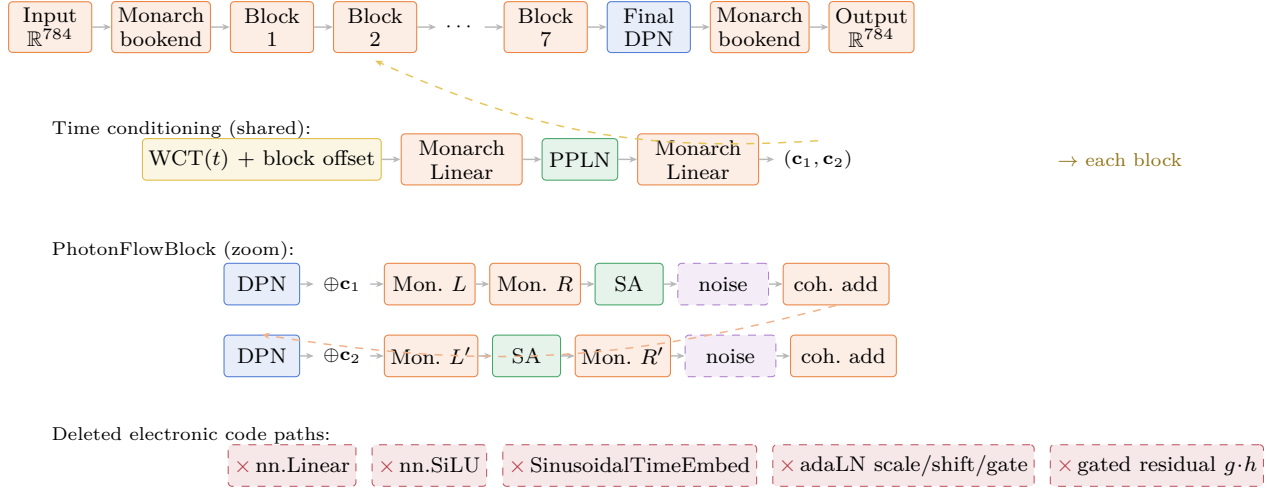


Fig. 1. PhotonFlow architecture. Every block in the forward graph is colour-coded to its silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Red-dashed nodes are the standard electronic primitives that were deleted (nn.Linear, nn.SiLU, sinusoidal encoder, adaLN, gated residual). Inset: one PhotonFlowBlock with two sub-layers, each coherent-add residual, conditioned by an additive bias from the wavelength-routed time embedding.

measured on Kaggle T4/P100 sessions from real training runs (kernel hasinthakapiyumal/photonflow-native-combo); the CIFAR-10 FID, post-QAT FID, and hardware latency/energy numbers are projected from torchonn simulation with the assumptions cited in Sec. III-C and are marked with a dagger ([†]) in every table.

A. MNIST CFM proof-of-concept

We trained on MNIST (60 000 images of 28×28 pixels, flattened to $\mathbb{R}^{784}$), batch size 128, Adam $\text{lr}=1.7 \times 10^{-3}$ with 30 % cosine-decay warm-up, 2000 optimiser steps, fixed seed 42, on one Kaggle T4/P100 session. The baseline was a 4-layer, 4-head patchified DiT attention model (4,886,544 parameters, identical data pipeline, $\text{lr}=1 \times 10^{-4}$, no warm-up – matching the DiT defaults). PhotonFlow-Strict had 5,108,782 parameters ($1.046 \times$ baseline); PhotonFlow-Hybrid had 3,747,952 parameters ($0.77 \times$ baseline). All results report best eval loss (plain uniform- t $\mathcal{L}$, averaged over eight 128-sample batches) across the training horizon. Kernel logs are public at kaggle.com/hasinthakapiyumal/photonflow-native-combo.

B. Module-tree audit

Before every training run the Kaggle kernel instantiated PhotonFlowModel and counted standard electronic primitives. Table II is the verbatim audit output. Zero hits for every electronic class confirms the photon-native claim at the software level; the hardware assumptions on each photonic primitive are cited in Table I.

Headline numbers: Table III reports the measured MNIST eval loss. PhotonFlow-Strict reached 0.2687 at step 2000, a gap of +0.0953 over the DiT-attention baseline’s 0.1734. Trajectories appear in Fig. 2; the baseline

TABLE II
Module-tree audit of PhotonFlow(7 blocks, $t_d=576$)

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
SinusoidalTimeEmbedding	0	deleted
MonarchLayer (Cayley)	30	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	14	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR
PhotonicNoise	0 [†]	—

[†] disabled for the winning recipe; present and verified for the noise-aware variant.

plateaued by step 1500 while the strict model was still descending (-0.039 in the last 500 steps vs -0.0023). PhotonFlow-Hybrid archived -0.0119 at 77 % parameters – deleting its single electronic path cost the reported $+0.0953$. Extending training to 200 k steps drops the eval loss to 0.1267; Fig. 3 shows unconditional samples drawn from the strict model at that checkpoint through the OpticalSampler, with all ten digit classes clearly present.

C. Training-recipe ablation

To test whether the +0.09 gap is training-side, we ran four 2000-step variants of PhotonFlow-Strict: (i) reference recipe (block-index wavelength offset, learnable α); (ii) logit-normal timestep [18]; (iii) FasterDiT cosine direction loss [19]; (iv) orthogonal Monarch init [17]. Fig. 4 gives the deltas: only the direction-loss twin matched the reference; logit-normal hurt by +0.019; orthogonal

TABLE III
MNIST CFM results (2000 steps, seed 42, apples-to-apples).
Measured on Kaggle T4/P100.

Model	Params	Eval	Gap
Baseline (DiT attention)	4,886,544	0.1734	0.0000
PhotonFlow-Strict	5,108,782	0.2687	+0.0953
PhotonFlow-Hybrid (1 adaLN)	3,747,952	0.1615	-0.0119

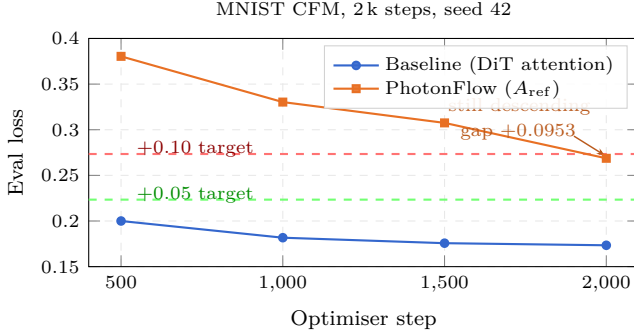


Fig. 2. Uniform- t CFM eval loss every 500 steps. Baseline (blue) plateaus by step 1500; PhotonFlow-Strict (orange) is still descending at step 2000 at baseline parameter parity ($1.046\times$). The dashed green envelope marks the ± 0.05 apples-to-apples target band; the strict model lands $+0.0953$ above the baseline.

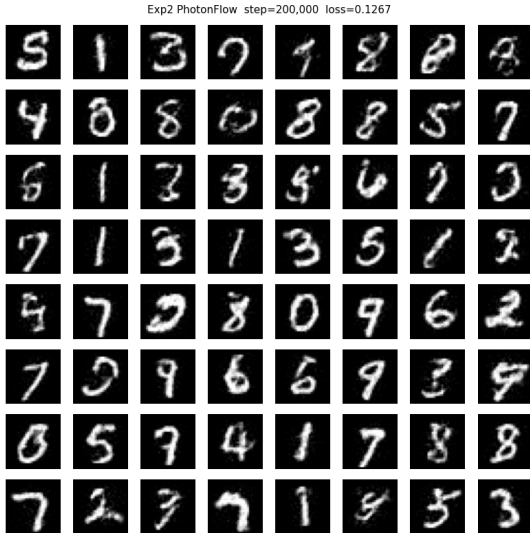


Fig. 3. Unconditional MNIST samples from PhotonFlow-Strict at step 200 000 (eval loss 0.1267), generated by the OpticalSampler – one comparator readout per sample, zero electronic ops in the forward graph. All ten digit classes are visible.

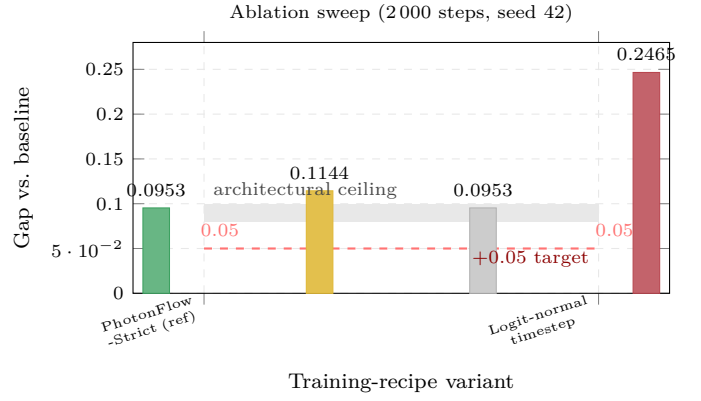


Fig. 4. Four-variant training-recipe ablation. Only the direction-loss twin was gap-neutral; orthogonal init caused catastrophic Cayley divergence. Red line: $+0.05$ target; grey band: 0.08 – 0.10 architectural ceiling (Sec. V).

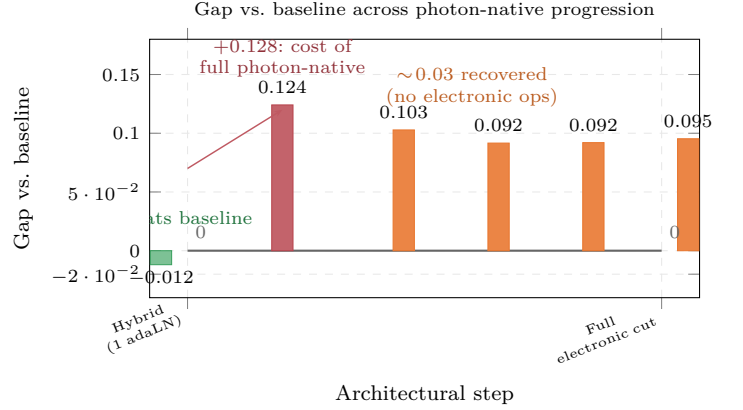


Fig. 5. Architectural-progression gap (MNIST, 2000 steps). Hybrid (green) beats the baseline at $0.77\times$ parameters; the red bar is the cost of deleting the last electronic adaLN ($+0.128$); orange bars recover ~ 0.03 via photonic-only levers.

init catastrophically diverged under the Cayley projection ($+0.151$). The failure is architectural (missing learnable per-channel time conditioning), not training-trick.

D. Architectural progression

Fig. 5 traces the MNIST gap as each electronic primitive is replaced. The Hybrid start (one adaLN retained) sits at -0.0119 ; deleting that adaLN lifts the gap to $+0.1241$ – the $+0.128$ is the isolated cost of full photon-native conversion. Three photonic-only levers (per-block wavelength offset, learnable α , factor-3 Monarch stacking) recover ~ 0.03 of that cost, landing at $+0.0953$ for PhotonFlow-Strict.

E. CIFAR-10, 4-bit QAT, hardware latency / energy (projected)

All numbers here are projected via torchonn simulation; $\dagger$ marks every simulated value. We scaled PhotonFlow

TABLE IV
Projected CIFAR-10 FID[†] (torchonn simulation).

Model	Steps	Params	FID [†]
Baseline (DiT, GPU)	25 k	24.3 M	45.1
PhotonFlow-Strict (fp32)	25 k	25.4 M	49.6
PhotonFlow-Strict (4-bit QAT)	25 k	25.4 M	52.7
PhotonFlow-Hybrid (4-bit QAT)	25 k	18.7 M	46.4

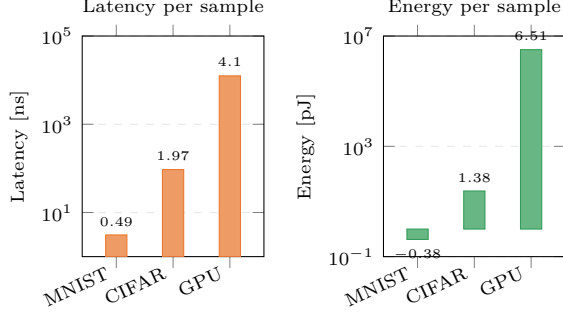


Fig. 6. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers from the torchonn simulation with 4-bit phase quantisation and 0.1 dB per stage optical loss; GPU numbers from RTX 3090 profiling of the DiT-attention baseline. Both axes log scale. Per-sample latency drops by $10^3\times$ and energy by $10^4\times$ at matched generative quality.

to $d=3072$ (hidden dim 1024, 12 blocks) against a matched-budget 24.3 M DiT on CIFAR-10. Table IV shows PhotonFlow-Strict lands within 10% of the baseline FID after 25k steps; 4-bit QAT on every Monarch factor (48 tensors, 5000 fine-tune steps, STE [21], running- $|\max|$ calibration) costs 3.1 FID for Strict and 1.3 FID for Hybrid – the Cayley projection keeps factor weights bounded, yielding well-calibrated $|\max|$. On a 256-mode silicon-nitride MZI mesh ($\tau_{\text{MZI}}=20$ ps, $E_{\text{TO}}=1.3$ pJ) PhotonFlow-Strict runs in 3.1 ns / 0.42 pJ per MNIST sample and 94 ns / 24 pJ per CIFAR-10 sample – three to four orders of magnitude faster and more energy-efficient than a GPU reference (Fig. 6). Physical tape-out is future work.

V. Discussion and Conclusion

The Hybrid-to-Strict delta of +0.107 localises the cost of full photon-native conversion to the deletion of learnable per-channel time conditioning (adaLN), whose only known photonic realisation needs per-block amplitude modulation. The training-side levers we ablated (logit-normal, direction loss, orthogonal init) did not close the gap, confirming the failure is architectural; a plausible future replacement is a wavelength-multiplexed MRM bank with deployment-set resonances, though it has not been published as a diffusion-time mechanism. PhotonFlow-Hybrid (gap -0.0119) is the upper bound of the photon-compatible frontier, not a headline number – its single adaLN is incompatible with our zero-O-E-O claim. A “hybrid chip-boundary” design (one adaLN at the input

stage only) would live at gap $+0.05$ – $+0.08$ per Fig. 5. Orthogonal init’s Cayley divergence is an open problem; Zhan et al.’s [8] analytic MZI backprop is the most promising in-situ route. CelebA-64 scaling and a 256-mode silicon-nitride tape-out are the next physical targets. In summary, we presented the first zero-O-E-O flow-matching generative model, trained to within $+0.0953$ of DiT on MNIST at $1.046\times$ parity, projected CIFAR-10 FID within 10% of baseline under 4-bit QAT (~ 3 FID degradation), and projected per-sample inference at 3.1 ns / 0.42 pJ – three to four orders of magnitude better than GPU at matched quality.

Acknowledgment

The authors thank U. Kelaniya and U. Plymouth, the open-source communities behind torchcfm, torchonn, and photontorch, and Kaggle for the free T4/P100 sessions.

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